

Static Timing Analysis

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Course Overview

1. STA Concepts
 - 1 lecture
2. Delay Modeling
 - 1 lecture
3. **Interconnect Parasitics**
 - 1 lecture
4. Delay Calculation
 - 1 lecture
5. Configuring the STA Environment
 - 1 lecture
6. Timing Checks
 - 1 lecture
7. Crosstalk and Noise
 - 1 lecture

3. Interconnect Parasitics

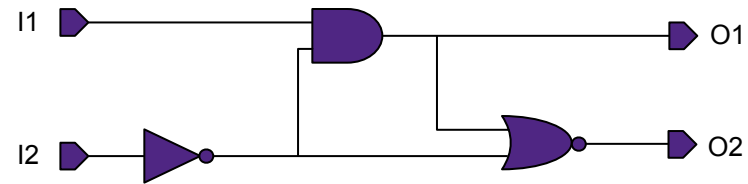
- **Net** is a wire connecting pins of standard cells and blocks. It:
 - Has only one driver
 - Can drive a number of fanout cells or blocks
 - Can travel on multiple metal layers of the chip
 - Can be broken up into segments for equivalent electrical representation

Interconnect Parasitics

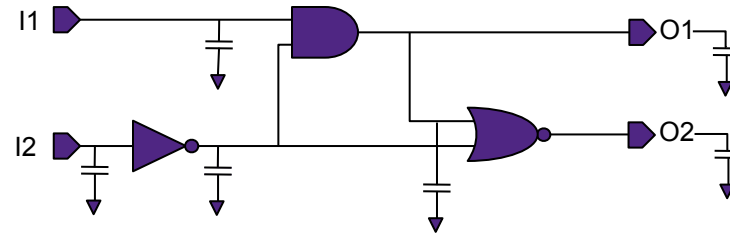
- **Interconnect resistance** - resistance between the output pin of a cell and the input pins of the fanout cells
- **Interconnect capacitance** - is comprised of grounded and between neighboring signal routes
- **Interconnect inductance** - arises due to current loops; effect of inductance can be ignored for low frequencies

Overview

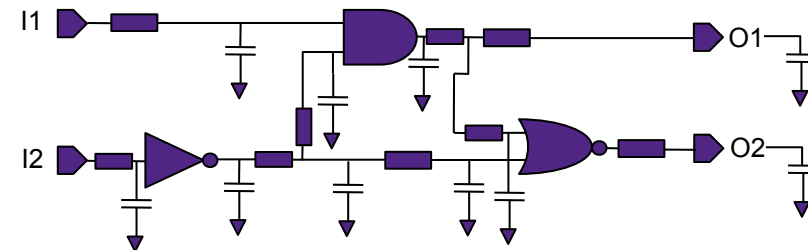
Cells



Input capacitances

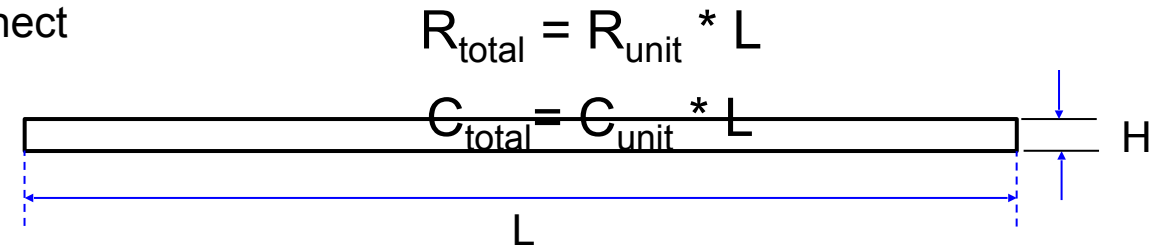


Net parasitics

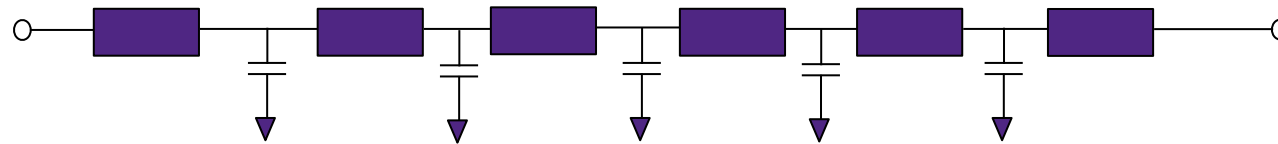


RC Tree of Interconnect

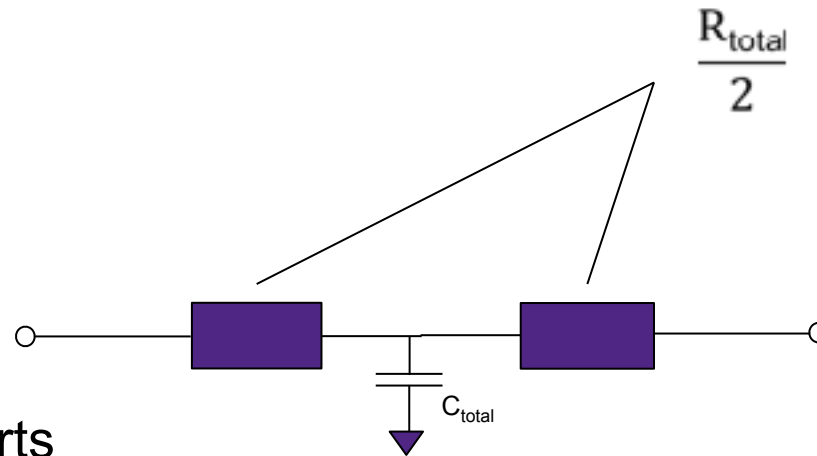
Interconnect



Distributed RC tree



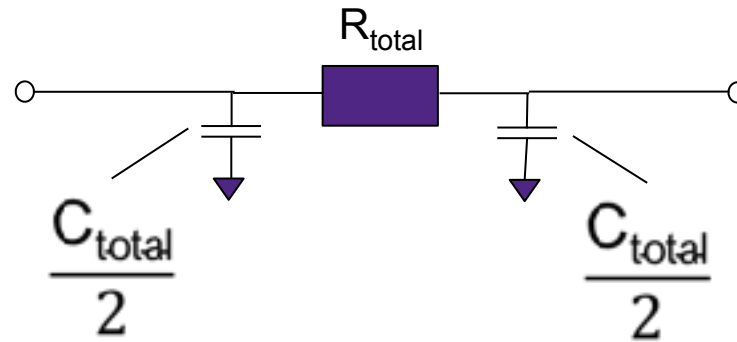
T - model



- R_{total} is broken in two parts
- C_{total} is connected between them

π - model

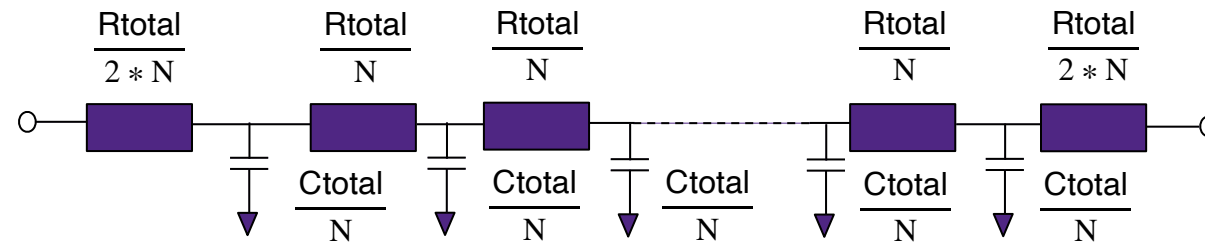
- C_{total} is broken into two parts
- R_{total} is connected between them



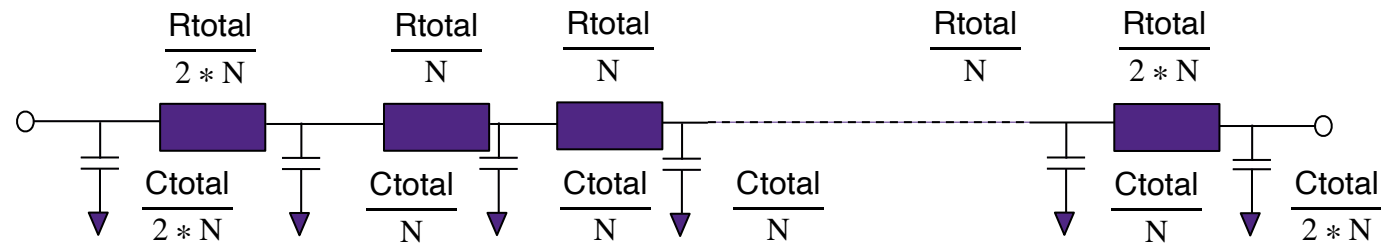
N Section Representation

- Breaking R_{total} and C_{total} into multiple sections increases accuracy

- T model:

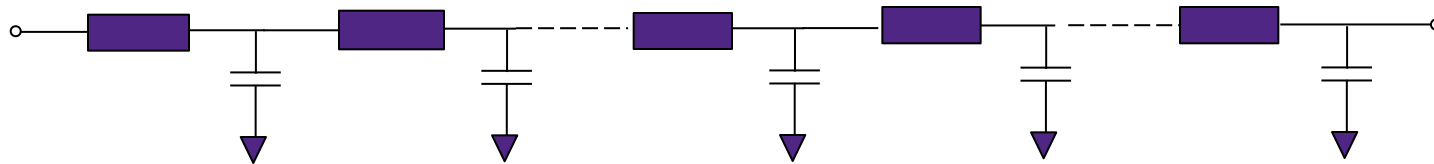


- π model:



Elmore Delay

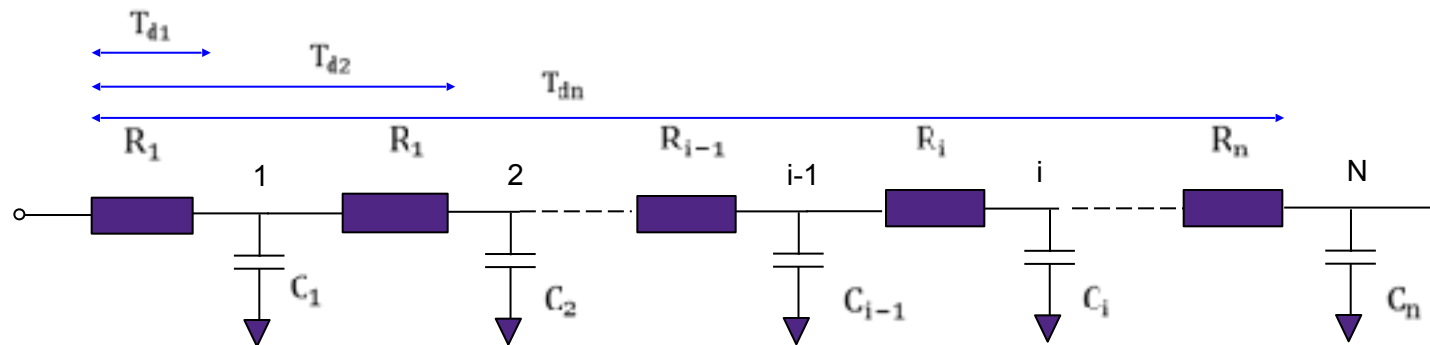
- **Elmore delays** are used to calculate delay of RC trees, which
 - Have single input node
 - Do not have resistive loops
 - Have capacitances only coupled to ground



Elmore Delay Equation

Elmore delay equation is:

$$T_{dn} = \sum_{i=1}^n \left[C_i \sum_{j=1}^i R_j \right]$$



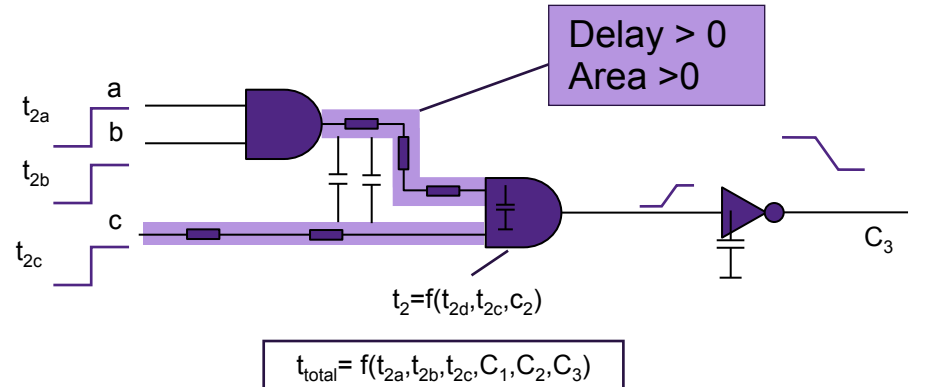
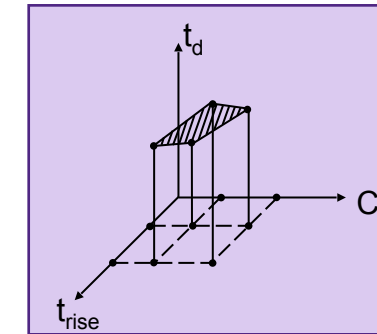
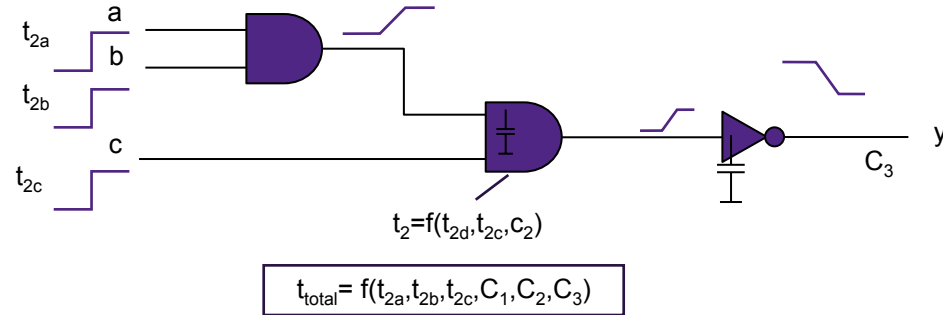
For example,

$$T_{d1} = C_1 * R_1$$

$$T_{d2} = C_1 * R_1 + C_2 * (R_1 + R_2)$$

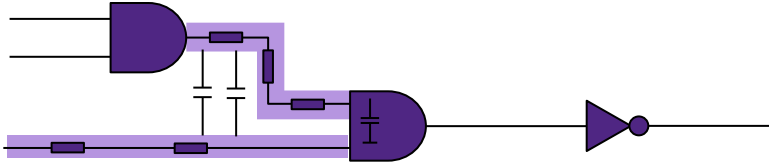
Wire Load: Parasitic Effects

- Parasitics are inevitable
- Not known without layout
- Delay and area will be incorrect/optimistic without estimation of parasitics



Estimating Parasitic Devices

Generalization



- All parasitics depend on interconnect length

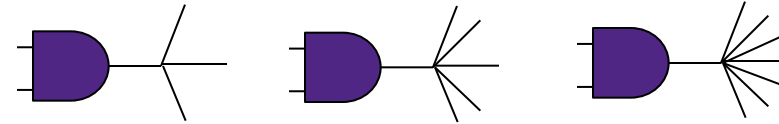
$$R = \text{length} \cdot R_{\text{unit length}}$$

$$C = \text{length} \cdot C_{\text{unit length}}$$

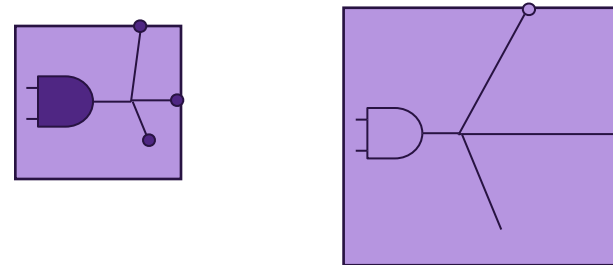
$$\text{Area} = \text{length} \cdot \text{Area}_{\text{unit length}}$$

Length components

- The more the fanout (output connections) the larger the length



- The larger the chip (the more gates it has) the more the length



$$\text{length} = f(\text{gate count, fanout})$$

Library: Wire Load Syntax

```
wire_load (name) {  
    area : float ;  
    capacitance : float ;  
    resistance : float ;  
    slope : float ;  
    {fanout_length (fanout, length) ;  
    ...  
}  
}  
wire_load_selection (name) {  
    {wire_load_from_area (min_area1,max_area1,wire_load_name1);  
    ...}  
}  
default_wire_load : name ;  
default_wire_load_selection : name ;  
default_wire_load_area : float ;  
default_wire_load_capacitance : float ;  
default_wire_load_resistance : float ;  
default_wire_load_mode : top | segmented | enclosed ;
```

Library: Wire Load Example

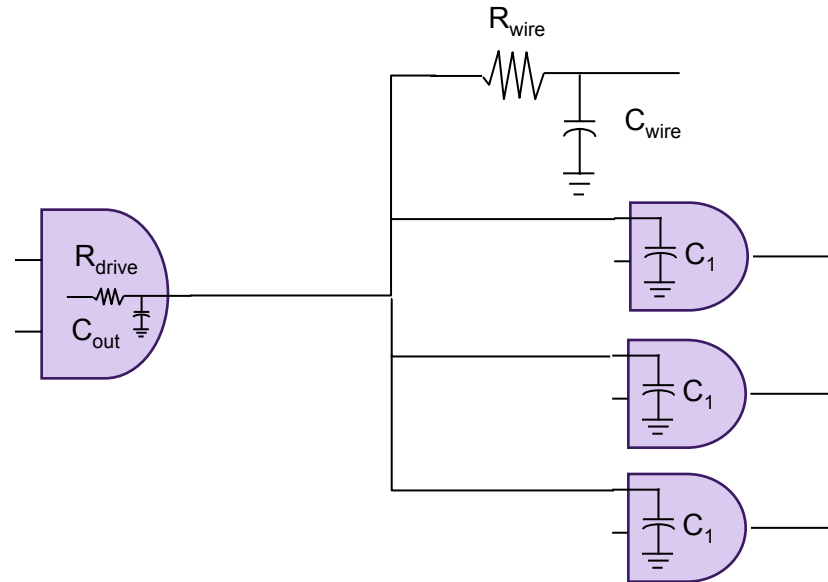
```
wire_load ("8000") {  
    capacitance : 0.024051;  
    resistance : 1.860000e-03;  
    area : 0.010000;  
    slope : 30.285426;  
    fanout_length("1", "8.2750360");  
    fanout_length("2", "18.4914880");  
    fanout_length("3", "29.3531220");  
}  
  
default_wire_load : "8000";  
default_wire_load_selection : "predcaps";  
  
wire_load_selection (predcaps) {  
    wire_load_from_area (0.000000,200.000000, "4000");  
    wire_load_from_area (200.000000,8000.000000, "8000");  
    wire_load_from_area (8000.000000,16000.000000, "16000");  
}  
}
```

Library: Wire Load Table

```
wire_load_table (name) {  
    fanout_length(fanout, length);  
    fanout_capacitance(fanout, capacitance);  
    fanout_resistance(fanout, resistance);  
    fanout_area(fanout, area);  
    ...  
}
```

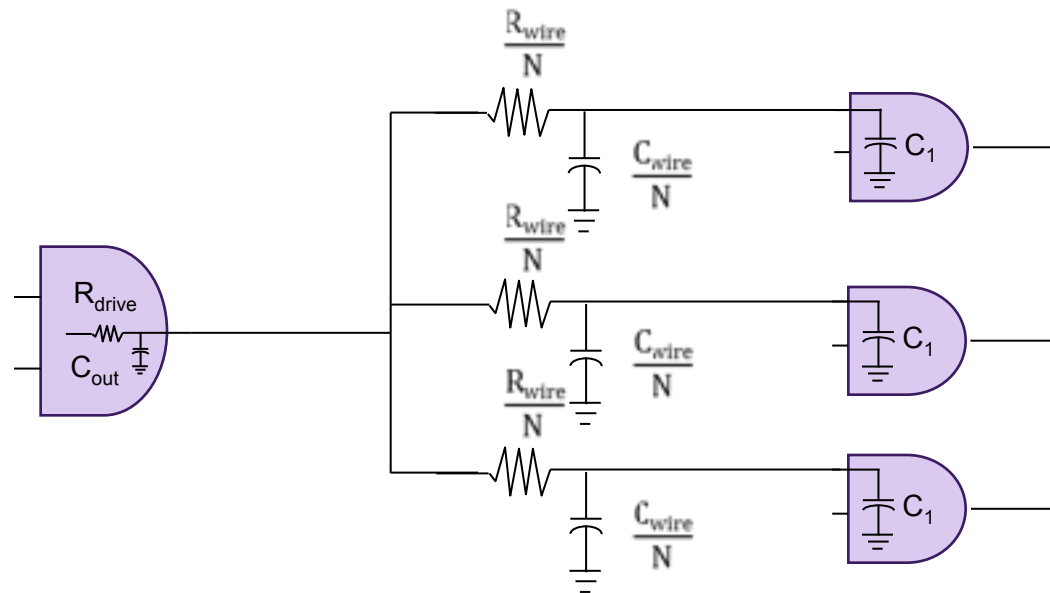
Interconnect Trees: Best-case Tree

- Assumed that:
 - Load pin is physically adjacent to the driver
 - None of the wire resistance is in the path to the destination pin
 - Wire capacitance and pin capacitances act as load on the driver pin



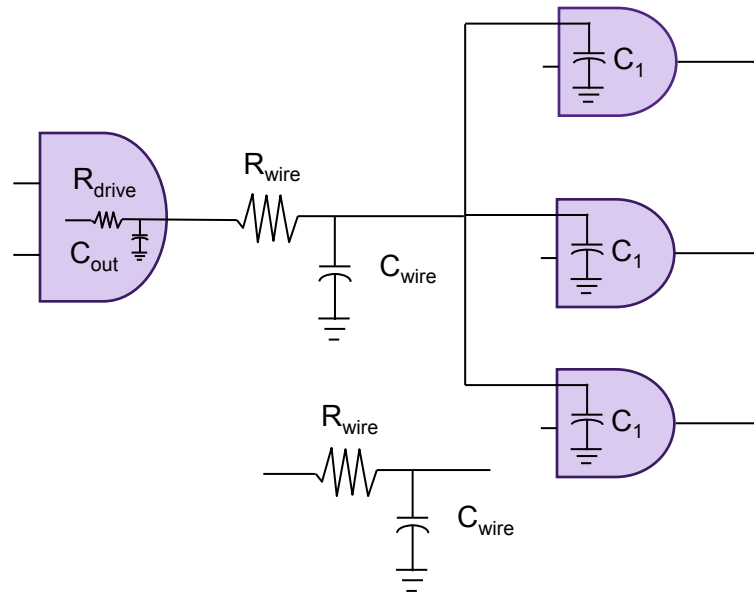
Interconnect Trees: Balanced Tree

- Assumed that:
 - Each destination pin is on a separate portion of the interconnect
 - Each path has equal portion of the total wire resistance and capacitance



Interconnect Trees: Worst-case Tree

- Assumed that:
 - All the destination pins are together at the far end of the wire
 - Each destination pin sees the total wire resistance and the total wire capacitance



Thank You